



The P/B-ROE Valuation Model

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To cite this article: Jarrold W. Wilcox (1984) The P/B-ROE Valuation Model, Financial Analysts Journal, 40:1, 58-66, DOI: [10.2469/faj.v40.n1.58](https://doi.org/10.2469/faj.v40.n1.58)

To link to this article: <https://doi.org/10.2469/faj.v40.n1.58>



Published online: 31 Dec 2018.



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The P/B-ROE Valuation Model

A valuation model based on expected growth in book equity, the P/B-ROE model is one of a wide variety of valuation models derivable from simple economic assumptions. It is useful in practice because historical return on equity is a fairly good indicator of future return on equity and because the statistics required to estimate the relationship are simple. Furthermore, the P/B-ROE model appears to be superior (in terms of its mean squared error) to price-earnings ratio models when applied to historical data for the majority of companies covered by the *Value Line Investment Survey*.

The P/B-ROE model allows estimation of investment horizon, required shareholder return and market consensus expected return on equity using historical data. Security analysts can thus compare their own estimates of future return on equity with the estimates of the market consensus. Their purchases should be biased toward stocks whose estimated average ROEs exceed the market consensus expected return on equity.

Finally, the P/B-ROE model leads to some interesting conclusions about the relations between stock price and earnings stability, dividend policy, leverage and beta. It suggests that stable earnings growth does not necessarily lead to higher prices, that dividends do matter, that leverage can be either good or bad, and that stocks with higher betas do not seem to have higher required returns in the way predicted by the Capital Asset Pricing Model.

A GOOD EQUITY VALUATION MODEL is (1) simple and understandable; (2) able to be tested and refined against historical data and precise in explaining current prices based on these data; (3) helpful to investors in isolating the market consensus estimate of future company performance; and (4) able to incorporate related ideas about financial variables and newly observed facts.¹

Figure A, based on a 1981 cross-section of food processing stocks, illustrates a strong linear relation between price-to-book value (P/B) (plotted on a logarithmic scale) and return on equity (ROE). One might be tempted to con-

1. Footnotes appear at end of article.

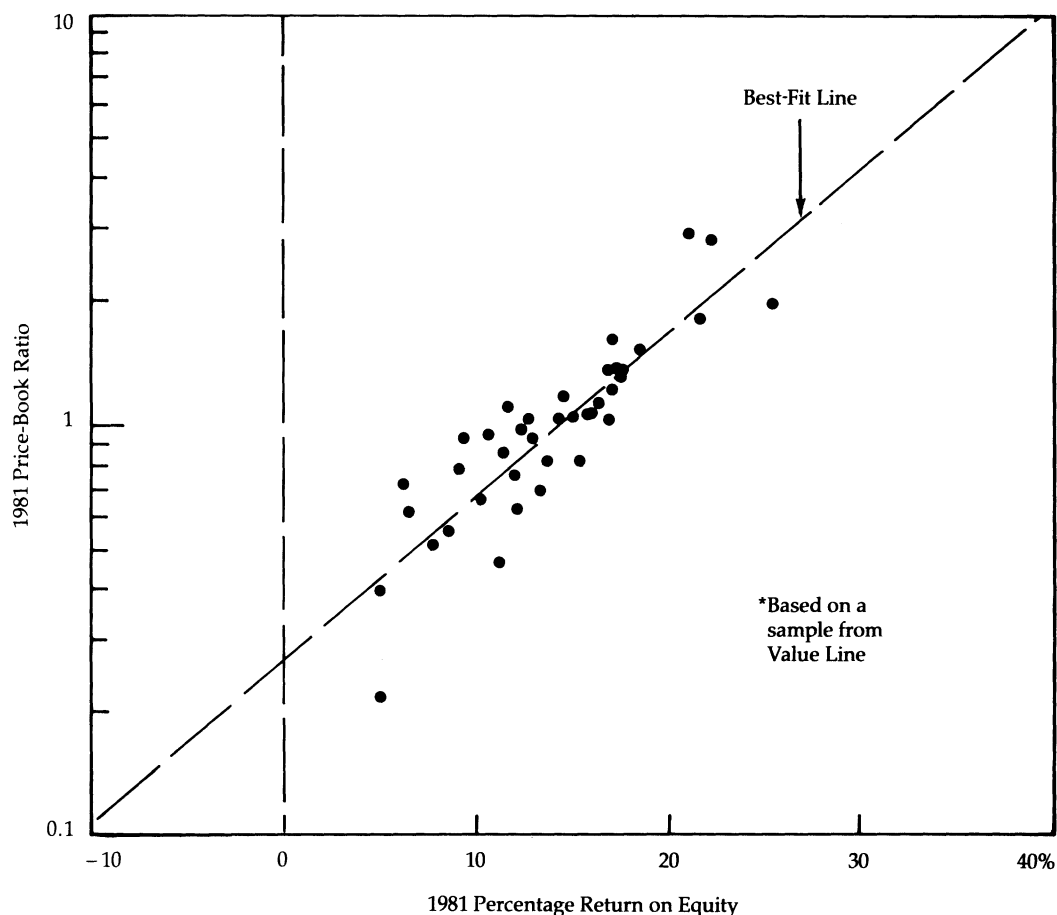
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clude that this relationship satisfies the criteria noted above. But does it have any basis in theory? Is it transferable to other industries and other points in time? Does it improve upon other valuation models, such as the traditional framework oriented around price-earnings ratios? The price-earnings ratio (P/E) is, after all, just (P/B)/ROE. How is statistical fit appraised for the P/B-ROE model when both sides of the equation involve division by the same variable—equity book value? Assuming one obtains a good fit, can the security analyst use it? Finally, can this rather simplistic relationship be used to explore empirically the impact of a wider range of financial variables? This article addresses each of these questions.

Theoretical Basis

By definition, realized shareholder holding period return equals dividend yield plus percentage

Figure A P/B and ROE for The Food Industry*



price change. Expected shareholder return equals expected yield plus expected percentage price change. Percentage price change can be meaningfully decomposed using a ratio of price to one of several performance descriptors—dividends, earnings, equity book value or even cash flow.²

In terms of equity book value, price is by definition:

$$P = B \cdot (P/B),$$

where P equals price and B book value. Price change is given by:³

$$\Delta P = \Delta B \cdot (P/B) + B \cdot \Delta(P/B) + \Delta B \cdot \Delta(P/B),$$

where

ΔP = change in price,

ΔB = change in book value and

$\Delta(P/B)$ = change in price-to-book-value ratio.

Percentage price change is obtained by dividing through by price:

$$\Delta P/P = \Delta B/B + \Delta(P/B)/(P/B) + \Delta B \cdot \Delta(P/B)/P.$$

In words, percentage price change equals percentage growth in book value, plus percentage change in the P/B multiple, plus a smaller term that combines both types of changes. The relationship is not causal; it is an inevitable consequence of the algebra.

Over a short enough time period, we may ignore the smaller, third term. Both actual and expected shareholder's return can then be viewed as yield plus book value growth plus percentage price-to-book ratio change.⁴ Labeling $\Delta B/B$ as g_B , the retained return on equity, we can express shareholder's return (k) as:

$$k = D/P + g_B + \frac{\Delta(P/B)}{P/B}, \quad (1)$$

Table I Alternative Compositions of Shareholder Return

Stock Type	Current Dividend Yield (D)	Retained ROE (g_B)	Change in P/B	Shareholder Return (k)
Electric Utility	10%	2%	2%	14%
Non-Dividend-Paying Industrial	0	14	0	14
Innovative Growth Firm, Leader in its Industry	4	18	-8	14
Break-Even Company	0	0	14	14

where D equals dividend yield on price. Table I gives some hypothetical examples of how Equation (1) works out for different types of stocks with equivalent shareholder returns.⁵

Henceforth, shareholder return, k , may be considered the weighted average expected return across investors (the market)—i.e., the required return. Competition fixes it so that, for stocks of similar risk, k is externally determined with respect to any single stock. Thus k is identical across broad classes of stock.⁶

An Equation for P/B

Equation (1) may be rearranged as follows:

$$P/B = \frac{D/B}{k - g_B} + \frac{\Delta(P/B)}{k - g_B}, \quad (2)$$

where D/B equals expected dividend yield on book value. The left-hand term suggests an infinite horizon dividend discount model. The information content of successive time periods, which must be summed up in the usual finite horizon dividend discount model, is represented here in the right-hand term, based on expected change in P/B over the first time period.

The continuous expected rate of change in P/B is given by:

$$\frac{d(P/B)}{dt} = P/B (k - g_B) - D/B. \quad (3)$$

When expected growth in book value and dividend yield on book are high enough, expected change in P/B must be negative. The market expects the P/B s of very profitable companies to decline.

Assume that investors expect return on equity to equal shareholder return at the end of an investment horizon and that price is expected eventually to equal book value (i.e., $P/B = 1$). Equation (3) governs the expected rate of change of P/B . Because P/B 's equilibrium value is set at unity, we can solve for current P/B ,

given an assumption of how we expect the performance parameters—growth in book value and dividend yield on book—to vary. If we assume, for simplicity, that both are constants throughout, then the exact solution for current P/B is:⁷

$$P/B = \frac{D/B}{k - g_B} + \left(\frac{k - r}{k - g_B} \right) e^{(g_B - k)T}, \quad (4)$$

where

- r = expected return on equity,
- T = investment horizon and
- e = the natural constant 2.718.

The implied price has the straightforward interpretation of being the sum of the present value of book value at the investment horizon plus the present value of dividends until then, assuming continuous compounding.⁸

In Equation (4), P/B depends slightly on the distribution of earnings between dividend yield on book and growth in book value. The larger effect is a combined dependence on r (expected return on equity). The present value formula for a dollar of book value continuously compounded at a rate r over T years with a time discount rate of k is a surprisingly useful approximation:

$$P/B = e^{(r-k)T}. \quad (5)$$

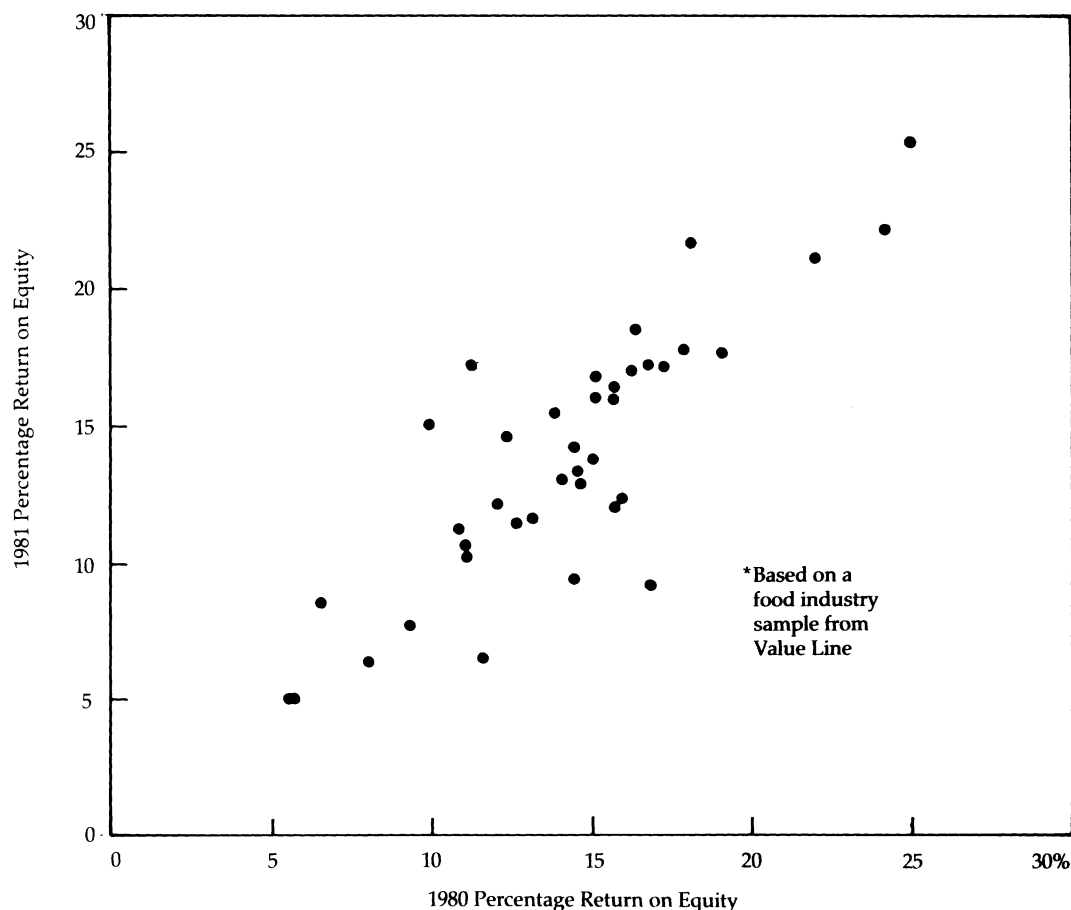
Equation (5) provides a simple log-linear model based on r :

$$\log(P/B) = -kT + Tr. \quad (6)$$

Historical return on equity (ROE) can frequently be substituted for expected return on equity (r) with only moderate weakening of the relationship. Figure B illustrates the ability of historical ROE to predict future ROE in a stable industry. In such cases, Equation (6) can be estimated using least-squares linear regression against historical accounting data.

Thus the simple P/B -ROE relationship ob-

Figure B Historical ROE as A Predictor of ROE*



served for the food processing industry does have a basis in theory. The results can be rigorously derived, as long as ROE is an empirical proxy for r . Unlike other performance measures, such as growth in earnings per share, ROE has the property of the past being a useful predictor of the future over a wide range of industries and classes of stock.⁹

Generality and Superiority of P/B-ROE

The relationship shown in Figure A does not hold for a series of points each representing the same stock at a different point in time. At a minimum, the required shareholder return may not be constant. Nor does the relationship work well for industries in which the past is a poor predictor. Thus the samples used in the study excluded companies with a five-year standard deviation in return on equity that exceeded

0.05.

The P/B-ROE relationship does hold cross-sectionally, however, although changes in k or T may cause variations. Figure C replicates Figure A for two additional industries.

Figure D demonstrates that the P/B-ROE relationship is useful even when industry differences are ignored. Here, year-end 1981 P/B ratios are plotted as a function of 1981 ROE for a sample of 1,013 Value Line stocks that had 1976-80 standard deviations of ROE less than 0.05. Most of the stocks are visually buried in the solid black areas near the best-fit least-squares regression line, for which a strong linear relationship exists.¹⁰

P/B-ROE vs. P/E

If, instead of P/B ratios, Figure D had plotted current price-earnings ratios, the results would

Figure C P/B and ROE Across Industries and Over Time

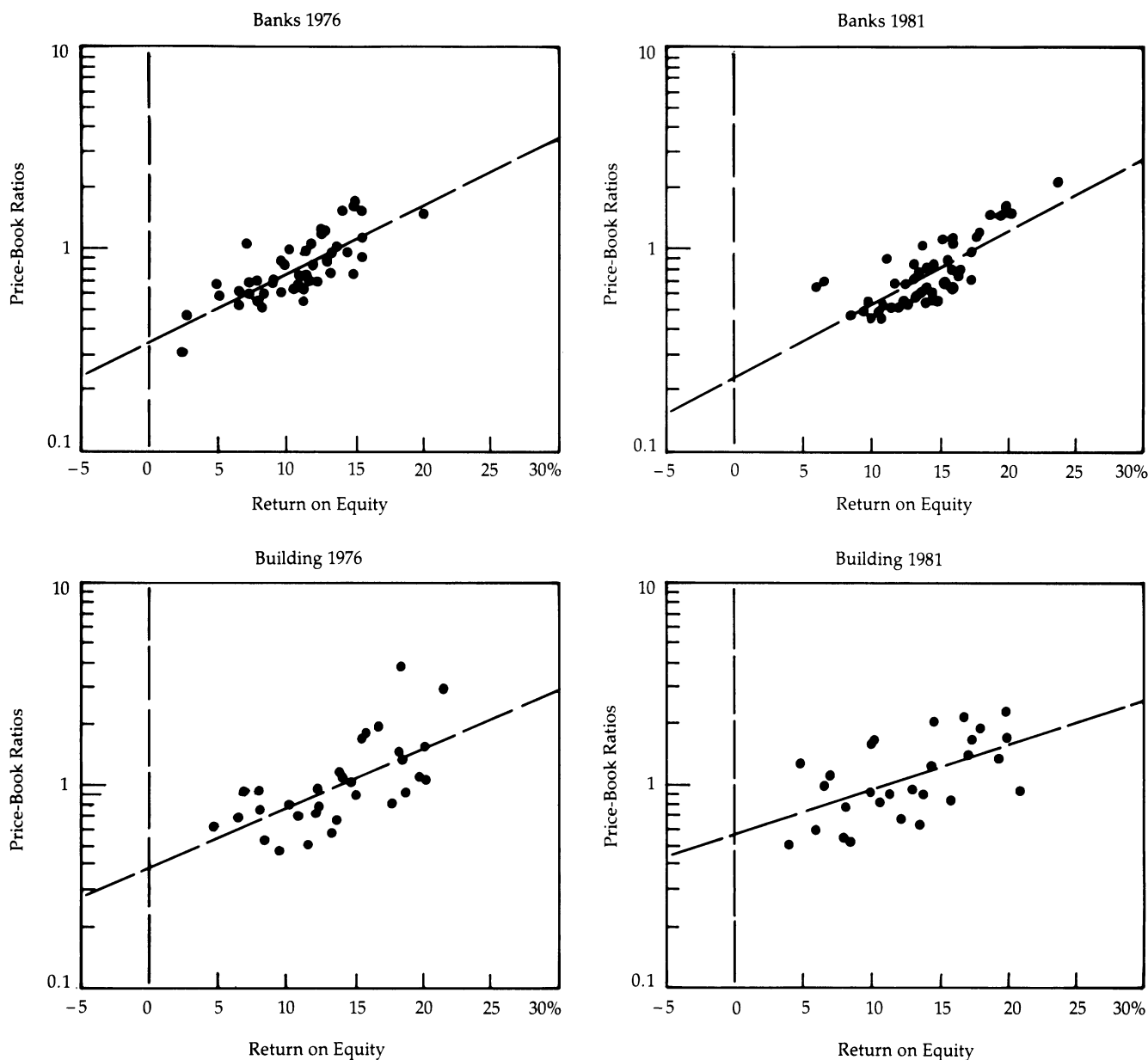


Table II The P/B-ROE Model vs. P/E Alternatives^a

Model	Mean of Squared Errors ^b	Standard Error ^c	Standard Deviation of Squared Errors
P/E = 9.90	0.239	0.489	0.742
P/three-year avg. E = 9.31	0.175	0.418	0.293
P/three-year avg. E = 7.45 + 14.36 (GE10) ^d	0.160	0.400	0.351
log P/B = 0.087	0.272	0.521	0.401
log P/B = -1.00 + 7.51 (ROE)	0.128	0.358	0.219

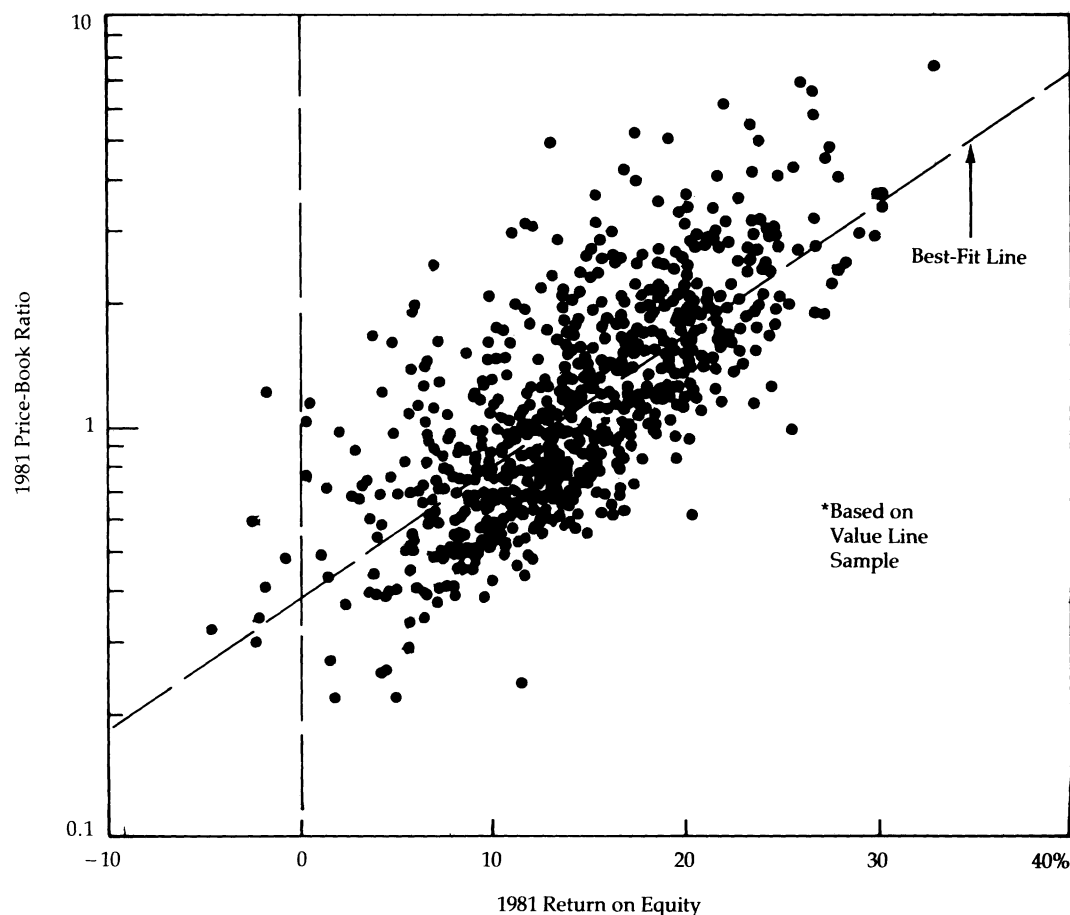
^a Based on a sample of 949 Value Line stocks for which all data are available, earnings are positive, and 1976–80 standard deviation of ROE is less than 0.05.

^b For each stock squared error = $\left[\log \left(\frac{\text{actual price multiple}}{\text{model price multiple}} \right) \right]^2$.

^c Standard error = square root of mean squared root.

^d Value Line 10-year growth in earnings.

Figure D P/B and ROE*



have exhibited no correlation! Furthermore, the negative earnings companies would not have appeared.

These difficulties are partly overcome by normalizing P/E, using a three-year average of earnings and Value Line's estimated 10-year growth rate in earnings per share as a trend line of earnings growth.¹¹ Using this approach, we begin to see some evidence of a positive relation between P/Es and historical earnings growth, as Figure E shows. Note, however, the counter tendency toward higher P/Es for very low growth rates.

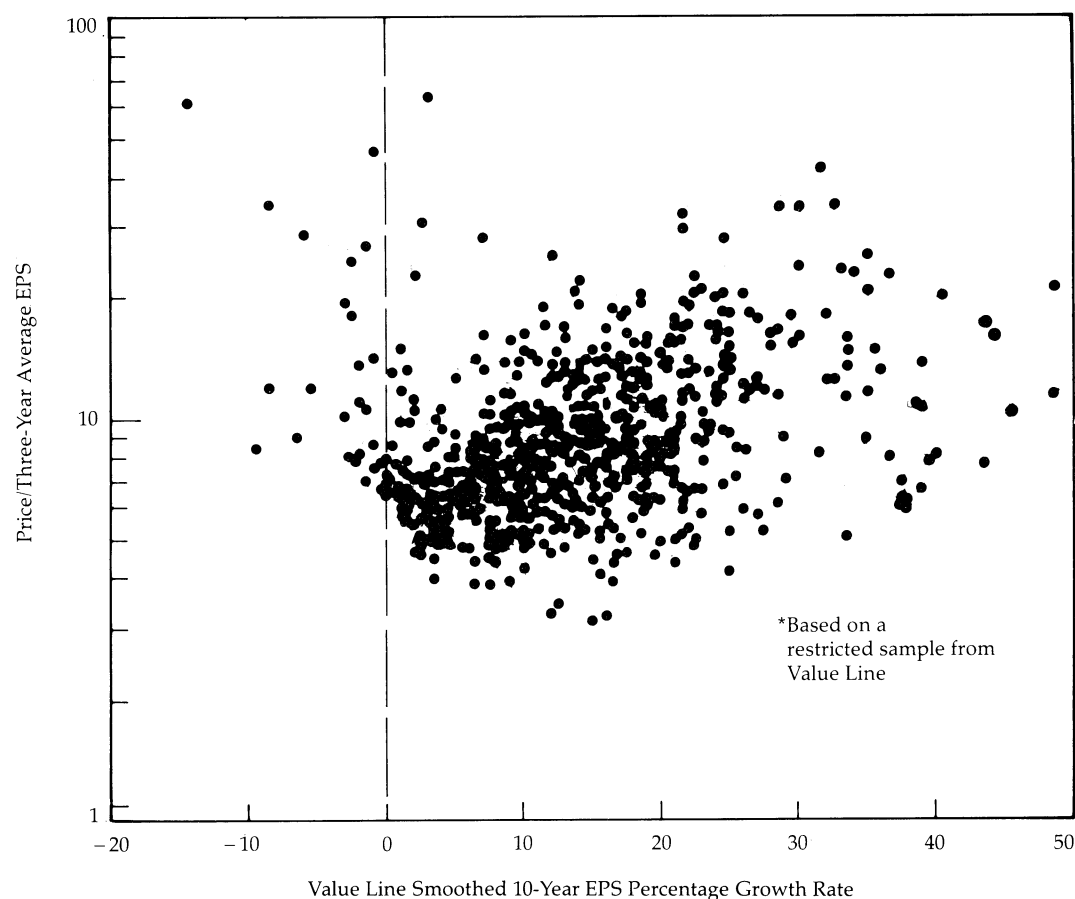
Does Figure D or Figure E provide the narrower band of expected price ratios? We cannot make the comparison based on R-squared, because the dependent variables are not the same, and B appears in both P/B and ROE (which is E/B). A better comparison is to analyze the mean squared errors, expressed as the log of the

ratio of actual price multiple to predicted price multiple. The errors are in each case equivalent to the log of the ratio of actual price to expected price.

Table II characterizes such errors for five simple models—(1) a constant P/E, (2) a constant P/three-year average E, (3) P/three-year average E as a linear function of 10-year earnings growth, (4) a constant P/B and (5) P/B as a log-linear function of ROE (Figure D). The standard error for a constant P/B is inferior to that for a constant P/E and much inferior to the standard errors for normalized P/E or a constant P/three-year average E.

These results help explain the predominant use of normalized P/E over P/B. When company performance is introduced, however, the roles are reversed. Adding 10-year historical earnings growth induces only a marginal improvement beyond the constant P/E. In contrast, the log P/B

Figure E 1981 Price-Earnings Ratios and 10-Year Earnings Growth*



versus ROE linear model has a standard error sharply lower than that of a constant P/B. The best standard error for a single variable P/E model was 0.400, which, translated as a positive percentage, is a miss of 49 per cent.¹² The standard error for the P/B-ROE model was 0.358, or 43 per cent—a noticeable improvement.¹³

Note also in Table II that the squared error varies much less across the sample for the P/B-ROE model than for the price-earnings models. Again, a linear P/E model is very far off the mark for low-earnings companies.

If the objective is to explain current prices based on historical data, the P/B-ROE model seems to offer a more accurate starting point than P/E models do. In addition, it is usable for companies with negative earnings and it requires only a single point estimate of the price

multiple.

In an industry by industry analysis, the P/B-ROE model seems to be at its best in explaining valuation differences between the stocks within a single industry, especially when performance histories are relatively stable and indicative of future expectations. The standard error in the food processing industry illustration (Figure A) is only 0.238. This corresponds to a typical positive miss of only 27 per cent, versus 43 per cent for the market as a whole.

The P/B-ROE Line

What is the meaning of the best-fit line relating P/B to ROE? Compare the diagonal line in Figure D to Equation (6):

$$\log P/B = -kT + Tr.$$

The equation of the best-fit line is:

$$\log P/B = a + b \text{ ROE}, \quad (7)$$

where a and b are the intercept and slope. As a first approximation, we take b as an estimator of T and $-a/b$ as an estimator of k .

Figure D's line of best fit is:

$$\log P/B = -0.974 + 7.41 \text{ ROE}, \quad (8)$$

giving for T , the estimated investment horizon, a value of 7.4 years and for k , the estimated required shareholder return, a value of 13.1 per cent ($0.974/7.41$). Note that k may be estimated by inspection of Figure D, because it is the ROE for which P/B equals unity.

Consensus Expected Return on Equity

An estimate of consensus expected return on equity for a security is obtained by observing the ROE on the P/B -ROE line in Figure D that corresponds to the observed P/B for the company in question. For many industries, however, one can improve substantially on this result by noting that the industry P/B -ROE line (such as that shown in Figure A) offers a better fit. It may more closely approximate the investment horizon characteristic of the stock in question, for example. Unfortunately, the intercept of the industry P/B -ROE line may be biased relative to that for the entire market. Although this bias may relate to differences in risk, it often occurs when expected return on equity for an industry appears to be expected to change relative to its history.

Average ROE may be near average r for the economy as a whole, but not necessarily for a single industry. For example, for the P/B -ROE line in Figure A, a least-squares regression gives:

$$\log P/B = -1.320 + 9.20 \text{ ROE}, \quad (9)$$

implying an estimated k of 14.3 per cent ($1.320/9.20$). A 1.2 per cent differential between this industry-relative k and the market k of 13.1 per cent may be viewed as an additional parameter—the industry average difference $\text{ROE} - r$. We may view $-a/b$ (derived from Equation (7)) when applied to industry best-fit lines as an estimator of $k + (\text{ROE} - r)$, rather than of k alone. That is, we might say that at year-end 1981, the market viewed the food industry's ROE as temporarily 1.2 per cent higher than its average value over the investment horizon. For many other industries, a larger absolute difference would have been observed.

Reiterating, one can obtain an estimate of market consensus expected return on equity (r) for an individual food industry stock as follows:

- (1) Calculate the ROE that, within the food industry, has the best fit with the company's actual P/B . From Equation (7), $\text{ROE} = (\log P/B + 1.320)/9.20$.
- (2) Subtract from this ROE the industry average differential ($\text{ROE} - r$), giving $r = (\log P/B + 1.320)/9.20 - 0.012$.
- (3) Substitute into this expression the stock's actual P/B . In the food industry at year-end 1981, a P/B of 2.0 implied an r of 20.7 per cent, and a P/B of 0.5 implied an r of 5.6 per cent.

If an analyst's own forecast average nine-year ROEs exceed these r s (nine years being the estimate of T for this industry), he or she should be biased toward purchase, and *vice versa*.

Clearly, operational security analysis will benefit from a price-value fit even tighter than the P/B -ROE line of Equation (7). Exploring a still more precise explanation of P/B differences is a task for the investment research profession. The chief practical milestones likely to be encountered in this pursuit will take into account: (1) proper assignment of industry effects; (2) dividend policy (Equation (4)); (3) intercompany differences in T , particularly as they relate to earnings variability; (4) impact of unusual risk on k ; (5) treatment of convertible issues; (6) interfirm differences in accounting methods; (7) the impact of inflation; and (8) additional sources of data, beyond current ROE, predictive of market consensus forecast return on equity.

Additional Observations

The P/B -ROE model leads to some interesting empirical findings. Among them are four that contradict commonly held beliefs; these are summarized below.

Stable earnings growth always leads to higher prices. This is not true. Stability of return on equity leads to a longer investment horizon. A longer horizon is associated with an increase in value when ROE is greater than k , but with a decrease in value when ROE is less than k . By splitting a large sample of stocks into contrasting halves of low and high ROE stability, one observes significant differences in the slope of the best-fit line.

Dividends don't matter. This is not true, but the effect is subtle. The difference between Equations (4) and (5) implies that dividends benefit

companies whose ROEs are less than k , but penalize companies whose ROEs are greater than k . Other valuation models make this prediction, but they offer little support in terms of actual measurement of the effect.

This dividend effect is apparent in the residuals about the P/B-ROE line. However, because dividend-paying companies are generally more stable and have longer investment horizons than non-dividend-paying companies, they may exhibit a price trend contrary to that predicted on the basis of dividend policy alone. When ROE stability is controlled for, dividend effect becomes more evident.

Leverage hurts. This is only sometimes true. The impact of debt leverage is extraordinarily variable in total. Incremental leverage can affect r , T and, in principle, k .

Stocks with higher betas have higher required return. The P/B-ROE model offers no evidence of this. No positive correlation exists between the k found in about 35 industry studies and the average beta within each industry. Also, no negative correlation is found between betas and the individual stock residuals around industry P/B-ROE best-fit lines.

Can the P/B-ROE approach be used in forecasting security betas? An inspection of Equation (6) suggests that price volatility will be dominated by:

$$(\Delta r - \Delta k)T + (r - k)\Delta T.$$

The left-hand term indicates that stocks with higher variation in r will have higher measured betas (given equal correlation with the market as a whole). But the right-hand term implies a tendency for both very high and very low P/Bs to be accompanied by higher betas. Furthermore, the net impact of the differences between Equations (4) and (5) is to flatten the slope of the P/B-ROE line for high-dividend companies. This will reduce beta (in the same way as a shorter investment horizon would) through an impact analogous to that of the left-hand term. High yield will be negatively correlated with security beta. This effect can be observed in practice. ■

Footnotes

1. For other views on practically useful valuation models, see Nicholas Molodovsky, "Stock Values and Stock Prices," *Financial Analysts Journal*, November/December 1968; Burton G. Malkiel and John G. Cragg, "Expectations and the Structure of Share Prices," *American Economic Review*, Sep-

tember 1970; Ralph A. Bing, "Stock Valuation: Theory vs. Practice," *Financial Analysts Journal*, May/June 1971; Joel Stern, "Earnings Per Share Don't Count," *Financial Analysts Journal*, July/August 1974; David F. Hawkins, "Toward an Old Theory of Equity Valuation," *Financial Analysts Journal*, November/December 1977; William L. Fouse, "Risk and Liquidity: The Keys to Stock Price Behavior," *Financial Analysts Journal*, May/June 1976; and William Beaver and Dale Morse, "What Determines Price-Earnings Ratios," *Financial Analysts Journal*, July/August 1978.

2. We show the method of decomposition using book value, but the same approach works for other variables. Normal earnings are used by Molodovsky ("Stock Values and Stock Prices," *op cit.*), cash flow by Stern ("Earnings Per Share Don't Count," *op cit.*) and dividends by Hawkins ("Toward an Old Theory of Equity Valuation," *op cit.*)
3. The derivation of this equation is available from the author on request.
4. Of course, for expected returns the quantities on the right are expected, not realized.
5. Similar analysis of price change using other performance descriptors leads to alternative perspectives on the components of expected shareholder return. The ideas presented in the text parallel to a remarkable extent those of Charles G. Callard in "The Third Yield," *Financial Analysts Journal*, January/February 1968.
6. Risk adjustments and portfolio theory are beyond the scope of this article. For examples of portfolio risk treatment, see William F. Sharpe, *Portfolio Theory and Capital Markets* (New York: McGraw-Hill, 1970). Attempts have been made to integrate valuation models and the resulting capital asset pricing model by Wells Fargo Bank and others.
7. That is, integrating over all the changes in P/B expected until the end of the investment horizon.
8. This is a derived result—one the author would not accept as a premise, because to do so would obscure its dependence on specific economic assumptions.
9. A. C. Paynor and I. M. D. Little, *Higgledy Piggledy Growth Again* (Oxford: Basil Blackwell, 1966). See also Guilford G. Babcock, "The Roots of Risk and Return," *Financial Analysts Journal*, January/February 1980, for a description of systematic versus unpredictable components of earnings growth.
10. The same would have been true if the figure had shown data for any year since 1959, which is as far back as I have checked.
11. This represents the best of a half-dozen single-variable P/E models tested.
12. That is, $(e^{0.40} - 1)$.
13. This improvement would have been even greater had I not bent over backward to favor P/E by excluding companies with negative ROEs.